

Analysis on Manifolds

May 16, 2026



Contents

1 Introduction 3

1.1 Smooth Curves 3

1.2 Path Integrals 6

1.3 Vector Fields and Line Integrals 6

1.4 The Reversed Parametrized Curve 8

1.5 Conservative Fields and Locally Conservative Fields 9

2 Smooth Manifolds in $\mathbb{R}^d$ 16

2.1 Transition Maps 19

2.2 Integrals On Parametrized Manifolds 19

1 Introduction

1.1 Smooth Curves

Definition 1.1

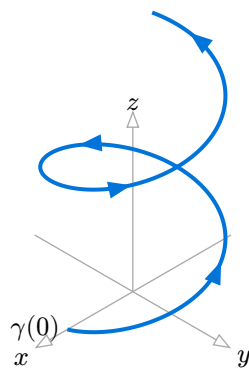
Parametrized Curve

A parametrized curve in $\mathbb{R}^n$ is a function $\gamma : [a, b] \rightarrow \mathbb{R}^n$ given by $\gamma(t) = (x_1(t), \dots, x_n(t))$ where each $x_i : [a, b] \rightarrow \mathbb{R}$. This defines a path in $\mathbb{R}^n$ that can describe movement over time.

Example

Helix

$\gamma : [0, 10] \rightarrow \mathbb{R}^3$ given by $\gamma(t) = (\cos(t), \sin(t), t)$ and the orientation is shown by the blue arrows in the figure.



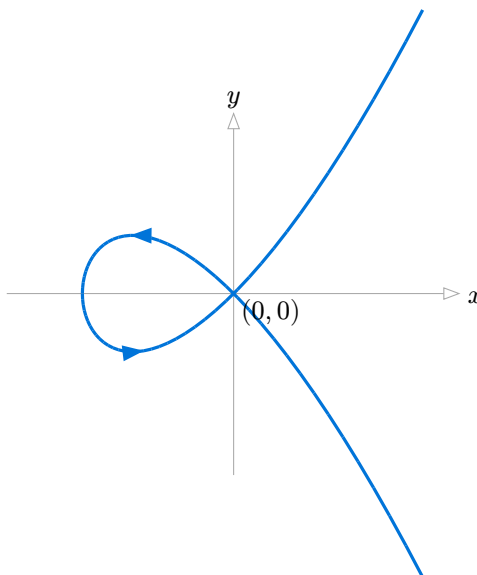
Example

Tschirnhausen cubic

$\gamma(t) = (t^2 - 1, t(t - 1)(t + 1)) = (t^2 - 1, t^3 - t)$ as t moves from $-\infty$ to ∞ in $\mathbb{R}^2$.

We notice that on the y axis we move through zero 3 times so this is a polynomial from degree 3 and the zeros of γ is at $t \in \{-1, 0, 1\}$ and on the x axis we go from ∞ to the point and back (this is a parabola).

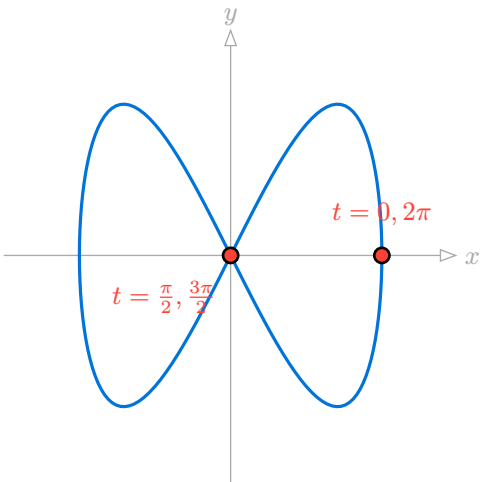
Observe that γ is a smooth parametrization curve but its image is not a smooth curve.



Example

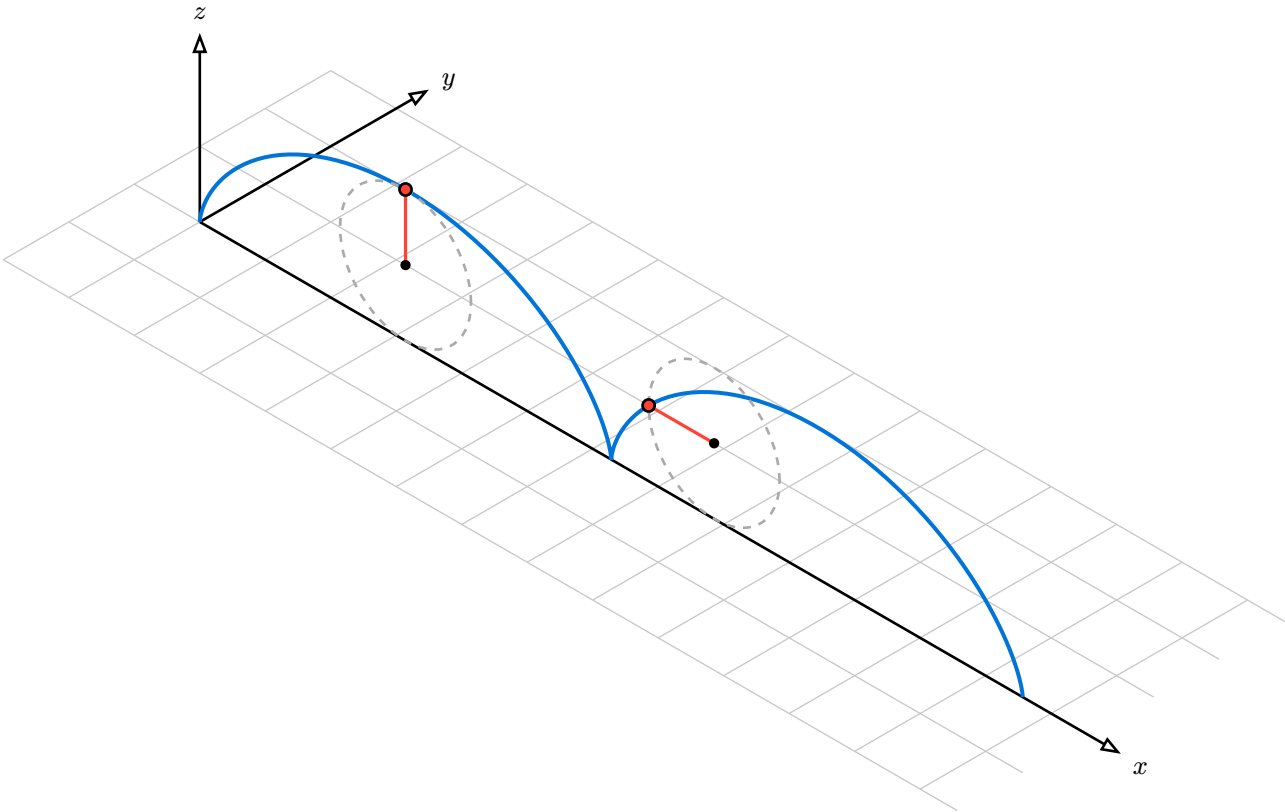
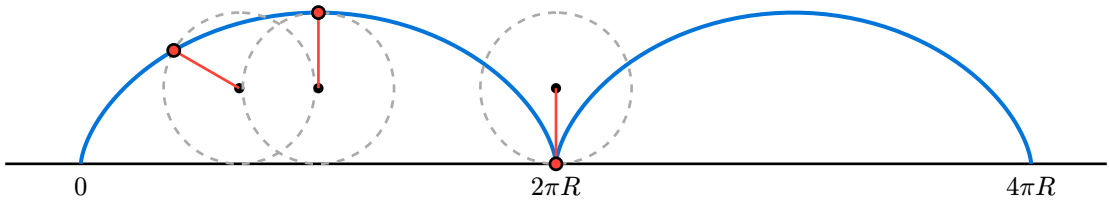
Lissajous Curve

$\gamma(t) = (\cos(t), \sin(2t))$



Example

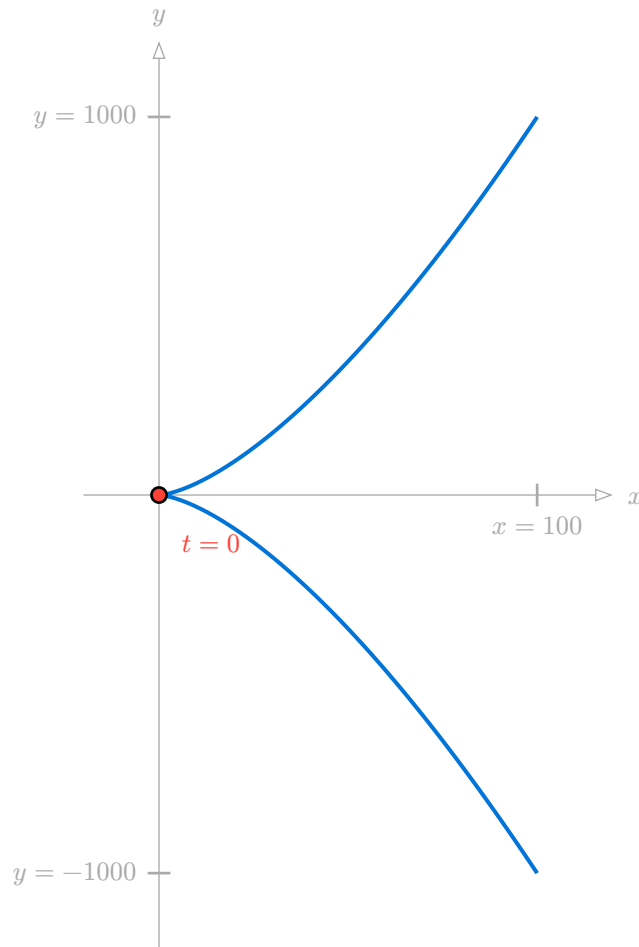
The Cycloid



Example

$\gamma : (-10, 10) \rightarrow \mathbb{R}^2$ given by $\gamma(t) = (t^2, t^3)$.

This is a smooth parametrization curve but not a smooth curve since $\dot{\gamma}(0) = (0, 0)$.



Definition 1.2

Velocity and Speed

Let $\gamma(t)$ be the position of a particle at time t . The **velocity** is the rate of change of position:

$$\gamma'(t) = \lim_{h \rightarrow 0} \frac{\gamma(t+h) - \gamma(t)}{h}$$

The vector $\gamma'(t)$ is the tangent to the curve at time t and the **speed** is the magnitude of the velocity: $\|\gamma'(t)\|$.

Definition 1.3

Regular Curve

A smooth parametrized curve is called **regular** if for all t , its speed is never zero: $\gamma'(t) \neq 0$.

Example

1. The curve $\gamma(t) = (t^2, t^3)$ is not regular because $\gamma'(0) = (0, 0)$
2. The cycloid $\gamma(\theta) = R(\theta - \sin \theta, 1 - \cos \theta)$ has $\gamma'(\theta) = R(1 - \cos \theta, \sin \theta) = (0, 0)$ when $\theta = 0, 2\pi, 4\pi, \dots$

1.2 Path Integrals

Definition 1.4

Length of Parametrized Curve

Let $\gamma : [a, b] \rightarrow \mathbb{R}^n$ a curve we denote the length of γ as $L(\gamma)$ by

$$L(\gamma) := \int_a^b \|\dot{\gamma}(t)\| dt$$

Definition 1.5

Path Integral (Scalar Field)

Let $\gamma : [a, b] \rightarrow \mathbb{R}^n$ be a smooth curve, and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ a continuous function. The path integral of f along γ with respect to the length element s is:

$$\int_{\gamma} f ds = \int_a^b f(\gamma(t)) \|\gamma'(t)\| dt$$

Example

If $f \equiv 1$, the path integral gives the length of the curve.

If γ represents a wire of non-uniform density, and $f(\gamma(t))$ is the density at a given point, then the path integral computes the total mass of the wire: $m = \int_{\gamma} f ds$.

Definition 1.6

Reparameterization

If $\varphi : [c, d] \rightarrow [a, b]$ is smooth, $\varphi(c) = a$, $\varphi(d) = b$, and $\varphi'(t) \neq 0$ (strictly increasing), then $\mu = \gamma \circ \varphi : [c, d] \rightarrow \mathbb{R}^n$ is a reparameterization of γ .

Lemma 1.7

Invariance of Path Integrals

The path integral $\int_{\gamma} f ds$ is well-defined, meaning it is invariant under reparameterization.

Proof: Let $\mu = \gamma \circ \varphi$. Using the chain rule, $\mu'(t) = \gamma'(\varphi(t))\varphi'(t)$.

Since $\varphi'(t) > 0$, $\|\mu'(t)\| = \|\gamma'(\varphi(t))\|\varphi'(t)$.

By substitution $\tau = \varphi(t)$:

$$\int_c^d f(\mu(t)) \|\mu'(t)\| dt = \int_c^d f(\gamma(\varphi(t))) \|\gamma'(\varphi(t))\| \varphi'(t) dt = \int_a^b f(\gamma(\tau)) \|\gamma'(\tau)\| d\tau = \int_{\gamma} f ds$$

□

End of Lecture 1 – 24/03

1.3 Vector Fields and Line Integrals

Definition 1.8

Vector Field

Let $\Omega \subseteq \mathbb{R}^n$ be a domain (connected, open set). A vector field on Ω is a smooth map $F : \Omega \rightarrow \mathbb{R}^n$.

Definition 1.9

Gradient Field and Potential

Let $u : \Omega \rightarrow \mathbb{R}$ be a smooth scalar function. Its **gradient field** is defined as $\vec{F} = \vec{\nabla}u = (\partial_1 u, \dots, \partial_n u)$.

The function u is called the **potential** of the field F .

Remark

Let $\hat{a} = (a_1, \dots, a_n)$ be a unit vector.

The directional derivative of u in the direction of $\hat{a}$ can be defined using a limit

$$\frac{\partial u}{\partial \hat{a}}(x) = \lim_{t \rightarrow 0} \frac{u(x + t\hat{a}) - u(x)}{t}$$

Notice that this is exactly the standard 1D derivative evaluated at $t = 0$

$$\frac{\partial u}{\partial \hat{a}}(x) = \frac{d}{dt} \Big|_{t=0} u(x + t\hat{a})$$

Applying the Chain Rule to expand this derivative, we get

$$\frac{d}{dt} \Big|_{t=0} u(x_1 + ta_1, \dots, x_n + ta_n) = \frac{\partial u}{\partial x_1}(x) \cdot a_1 + \dots + \frac{\partial u}{\partial x_n}(x) \cdot a_n$$

This sum is precisely the inner product of the gradient and the direction vector

$$\frac{\partial u}{\partial \hat{a}} = \langle \nabla u(x), \hat{a} \rangle$$

Now, taking the absolute value and applying the Cauchy-Schwarz inequality:

$$\left| \frac{\partial u}{\partial \hat{a}} \right| = |\langle \nabla u, \hat{a} \rangle| \leq \|\nabla u\| \|\hat{a}\|$$

Since $\hat{a}$ is a unit vector ($\|\hat{a}\| = 1$), this simplifies to $\left| \frac{\partial u}{\partial \hat{a}} \right| \leq \|\nabla u\|$.

Hence, the maximum directional derivative is obtained when $\hat{a}$ points in the exact same direction as the gradient, which is when

$$\hat{a} = \frac{\nabla u}{\|\nabla u\|}$$

End of Lecture 2 – 26/03

Definition 1.10

Line Integral

Let $\Omega \subseteq \mathbb{R}^d$ be a domain (open, connected) and let $F : \Omega \rightarrow \mathbb{R}^d$ be a vector field.

Let $\gamma : [a, b] \rightarrow \Omega$ be a parametrized curve.

The line integral is defined as:

$$\int_{\gamma} \vec{F} d\vec{t} = \int_a^b \langle F(\gamma(t)), \gamma'(t) \rangle dt$$

If F is a force field, then this integral computes the work done by the field on a particle moving along γ .

Example

1. Let $F_1(x, y) = (x, y)$ and let $\gamma_1(t) = (R \cos t, R \sin t)$ for $0 \leq t \leq 2\pi$

$$\int_{\gamma_1} \vec{F}_1 d\vec{t} = \int_0^{2\pi} \left\langle \begin{pmatrix} R \cos t \\ R \sin t \end{pmatrix}, \begin{pmatrix} -R \sin t \\ R \cos t \end{pmatrix} \right\rangle dt = 0$$

2. Let $F_2(x, y) = (-y, x)$ and let γ_1 be the same curve as before

$$\int_{\gamma_1} \vec{F}_2 d\vec{t} = \int_0^{2\pi} \left\langle \begin{pmatrix} -R \sin t \\ R \cos t \end{pmatrix}, \begin{pmatrix} -R \sin t \\ R \cos t \end{pmatrix} \right\rangle dt = \int_0^{2\pi} R^2 dt = 2\pi R^2$$

3. Let $F_2(x, y) = (-y, x)$ as before, but this time we integrate over the curve $\gamma_2(t) = (R - t, 0)$ for $0 \leq t \leq 2R$.

The derivative is $\gamma_2'(t) = (-1, 0)$, and $F_2(\gamma_2(t)) = F_2(R - t, 0) = (0, R - t)$

$$\int_{\gamma_2} \vec{F}_2 d\vec{t} = \int_0^{2R} \left\langle \begin{pmatrix} 0 \\ R - t \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\rangle dt = 0$$

Remark

Observe that both γ_1 and γ_2 start and end at the exact same points, yet, we don't have the same result for the integral ($\pi R^2 \neq 0$).

Lemma 1.11

Invariance under Reparameterization

The line integral $\int_{\gamma} \vec{F} d\vec{t}$ is well-defined, i.e., it's invariant under reparameterization of γ

Proof: Let $\varphi : [c, d] \rightarrow [a, b]$ be a reparameterization and define $\mu = \gamma \circ \varphi : [c, d] \rightarrow \Omega$ so we compute

$$\begin{aligned} \int_{\mu} \vec{F} d\vec{t} &= \int_c^d \langle F(\mu(s)), \mu'(s) \rangle ds \\ &= \int_c^d \langle F(\gamma(\varphi(s))), (\gamma \circ \varphi)'(s) \rangle ds \\ &= \int_c^d \langle F(\gamma(\varphi(s))), \gamma'(\varphi(s)) \cdot \underbrace{\varphi'(s)}_{\text{scalar}} \rangle ds \\ &= \int_c^d \langle F(\gamma(\varphi(s))), \gamma'(\varphi(s)) \rangle \varphi'(s) ds \end{aligned}$$

By using the substitution $t = \varphi(s)$, we obtain:

$$\int_a^b \langle F(\gamma(t)), \gamma'(t) \rangle dt = \int_{\gamma} \vec{F} d\vec{t}$$

□

1.4 The Reversed Parametrized Curve

Definition 1.12

Reversed Curve

Given a curve $\gamma : [a, b] \rightarrow \Omega$, the reversed curve $-\gamma$ has the exact same image, just traversed in the opposite direction. It is defined by $-\gamma(t) = \gamma(a + b - t)$ for $t \in [a, b]$

Lemma 1.13

Integration over Reversed Curves

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $F : \Omega \rightarrow \mathbb{R}^d$

1. $\int_{-\gamma} f ds = \int_{\gamma} f ds$
2. $\int_{-\gamma} \vec{F} d\vec{t} = - \int_{\gamma} \vec{F} d\vec{t}$

Proof:

$$1. \quad \int_{-\gamma} f ds = \int_a^b f(-\gamma(t)) \|-\gamma'(t)\| dt = \int_a^b f(\gamma(a + b - t)) \|-\gamma'(a + b - t)\| dt$$

Substitute $s = a + b - t$, so $ds = -dt$. The bounds change from $a \rightarrow b$ to $b \rightarrow a$:

$$\int_b^a f(\gamma(s)) \|\gamma'(s)\| (-ds) = \int_a^b f(\gamma(s)) \|\gamma'(s)\| ds = \int_{\gamma} f ds$$

$$2. \quad \int_{-\gamma} \vec{F} d\vec{t} = \int_a^b \langle F(-\gamma(t)), -\gamma'(t) \rangle dt = \int_a^b \langle F(\gamma(a + b - t)), -\gamma'(a + b - t) \rangle dt$$

Using the same substitution $s = a + b - t$ and $ds = -dt$:

$$\int_b^a \langle F(\gamma(s)), -\gamma'(s) \rangle (-ds) = - \int_a^b \langle F(\gamma(s)), \gamma'(s) \rangle ds = - \int_{\gamma} \vec{F} d\vec{t}$$

□

1.5 Conservative Fields and Locally Conservative Fields

Theorem 1.14

Gradient Fields Evaluate to Endpoints

Let $F : \Omega \rightarrow \mathbb{R}^d$ be a gradient field with potential $u : \Omega \rightarrow \mathbb{R}$ (i.e., $F = \nabla u$). Then $\int_{\gamma} \vec{F} d\vec{t} = u(\gamma(b)) - u(\gamma(a))$.

Proof:

$$\int_a^b \langle (\nabla u)(\gamma(t)), \gamma'(t) \rangle dt = \int_a^b ((\partial_1 u)(\gamma(t))\gamma'_1(t) + \dots + (\partial_d u)(\gamma(t))\gamma'_d(t)) dt = \int_a^b \frac{d}{dt}(u \circ \gamma)(t) dt = u(\gamma(b)) - u(\gamma(a))$$

□

Definition 1.15

Conservative Field

A vector field $F : \Omega \rightarrow \mathbb{R}^d$ is called a **conservative field** if for all curves $\gamma : [a, b] \rightarrow \Omega$, the integral $\int_{\gamma} \vec{F} d\vec{t}$ depends only on the endpoints of γ .

Example

Every gradient field is conservative (from the theorem above).

Lemma 1.16

Closed Curves

F is conservative if and only if for all closed curves $\gamma : [a, b] \rightarrow \Omega$ (i.e., $\gamma(a) = \gamma(b)$), we have $\int_{\gamma} \vec{F} d\vec{t} = 0$.

Proof:

$\Rightarrow$ Let $\gamma : [a, b] \rightarrow \Omega$ be a closed curve. Let ξ be the constant curve $\xi(t) = \gamma(a)$.

Since both curves share the same start and end points, and F is conservative, their integrals must be equal

$$\int_{\gamma} \vec{F} d\vec{t} = \int_{\xi} \vec{F} d\vec{t} = 0$$

$\Leftarrow$ Assume the integral over any closed loop is zero. If γ and μ share endpoints, the curve $\gamma - \mu$ (concatenation of γ and $-\mu$) is closed

$$0 = \int_{\gamma - \mu} \vec{F} d\vec{t} = \int_{\gamma} \vec{F} d\vec{t} + \int_{-\mu} \vec{F} d\vec{t} = \int_{\gamma} \vec{F} d\vec{t} - \int_{\mu} \vec{F} d\vec{t}$$

Thus, $\int_{\gamma} = \int_{\mu}$, meaning it depends only on the endpoints. □

Theorem 1.17

Conservative iff Gradient Field

Let Ω be a domain. F is conservative in Ω if and only if F is a gradient field in Ω .

Proof:

$\Leftarrow$ We already proved that every gradient field is conservative.

$\Rightarrow$ Assume F is conservative. We define a potential $u(x)$ by:

$$u(x) = \int_{x_0 \rightarrow x} \vec{F} d\vec{t}$$

where we choose any C^1 parametrized curve connecting x_0 to x in Ω .

We notice that $u(x)$ is well-defined (does not depend on the specific curve chosen), since F is conservative and the partial derivative is

$$\lim_{h \rightarrow 0} \frac{u(x + he_k) - u(x)}{h} = \frac{1}{h} \left(\int_{x \rightarrow x + he_k} \vec{F} d\vec{t} \right)$$

We parametrize the path as $\gamma(t) = x + te_k$ for $0 < t \leq h$.

$$= \frac{1}{h} \int_0^h \langle F(x + te_k), e_k \rangle dt = \frac{1}{h} \int_0^h F_k(x + te_k) dt$$

By the Fundamental Theorem of Calculus, if we write $f(t) = F_k(x + te_k)$ and $I(h) = \int_0^h f(t) dt$, then:

$$\lim_{h \rightarrow 0} \frac{1}{h} \int_0^h F_k(x + te_k) dt = I'(0) = f(0) = F_k(x)$$

What we get is that $\partial_k u(x) = F_k(x)$ for all k , which implies $\nabla u = F$ □

End of Lecture 3 – 14/04

Corollary 1.18

Locally Conservative Condition

Let $F = (F_1(x), \dots, F_d(x)) : \Omega \rightarrow \mathbb{R}^d$ be a conservative field. Then for all $1 \leq k, \ell \leq d$:

$$\partial_k F_\ell = \partial_\ell F_k$$

Is the converse true? the answer is No.

Example

Consider the vector field given by $F = (Im(\frac{1}{z}), Re(\frac{1}{z}))$ on the domain $\Omega = \mathbb{R}^2 \setminus \{(0, 0)\}$.

We notice that

$$\frac{1}{z} = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}$$

So we have

$$F(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$$

Let's check if $\partial_y F_1 = \partial_x F_2$:

$$\begin{aligned} \partial_y F_1 &= \partial_y \left(-y(x^2 + y^2)^{-1} \right) = -(x^2 + y^2)^{-1} - y \left(-(x^2 + y^2)^{-2} \cdot 2y \right) = \frac{-(x^2 + y^2) + 2y^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} \\ \partial_x F_2 &= \partial_x \left(x(x^2 + y^2)^{-1} \right) = (x^2 + y^2)^{-1} + x \left(-(x^2 + y^2)^{-2} \cdot 2x \right) = \frac{x^2 + y^2 - 2x^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} \end{aligned}$$

Thus, $\partial_1 F_2 = \partial_2 F_1$, satisfying the locally conservative condition.

We claim that F is not conservative in $\mathbb{R}^2 \setminus \{(0, 0)\}$.

Let $\gamma(t) = (R \cos t, R \sin t)$ for $0 \leq t \leq 2\pi$ so

$$\begin{aligned} \int_\gamma \vec{F} d\vec{t} &= \int_0^{2\pi} \langle F(\gamma(t)), \gamma'(t) \rangle dt \\ &= \frac{1}{R^2} \int_0^{2\pi} \left\langle \begin{pmatrix} -R \sin t \\ R \cos t \end{pmatrix}, \begin{pmatrix} -R \sin t \\ R \cos t \end{pmatrix} \right\rangle dt \\ &= \int_0^{2\pi} \left(\frac{R^2 \sin^2 t + R^2 \cos^2 t}{R^2} \right) dt \\ &= \int_0^{2\pi} 1 dt = 2\pi \neq 0 \end{aligned}$$

Notice that the result is 2π , which is independent of R , but since the integral over a closed loop is not zero, F is not conservative.

Remark

Let us try to find a potential for F by brute force. We want $\partial_x u = \frac{-y}{x^2 + y^2}$ and $\partial_y u = \frac{x}{x^2 + y^2}$ and integrating these yields

$$u(x, y) = \arctan\left(\frac{y}{x}\right)$$

However, this potential function is not defined globally on $\mathbb{R}^2 \setminus \{(0, 0)\}$; it is only defined on an open subset (where $x \neq 0$).

Theorem 1.19**Poincaré Lemma for Balls**

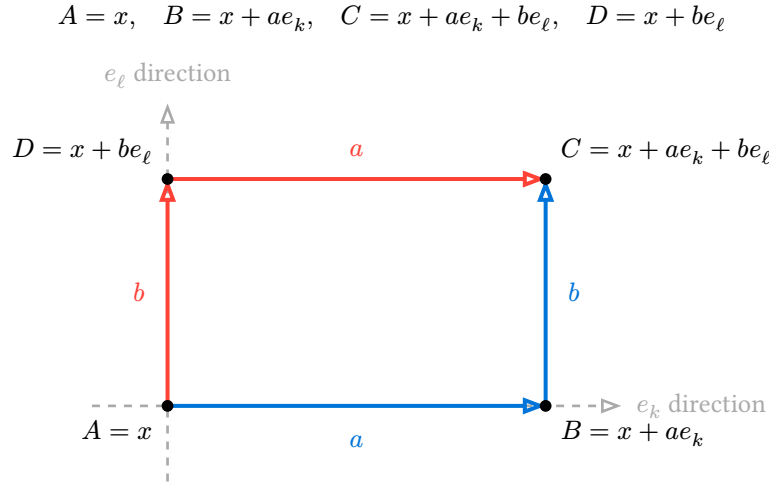
Let B be an open ball in $\mathbb{R}^d$. If $F : B \rightarrow \mathbb{R}^d$ satisfies $\partial_k F_\ell = \partial_\ell F_k$ for all $1 \leq k, \ell \leq d$, then F is conservative in B .

Theorem 1.20**Poincaré Lemma for Balls**

Let $B \subseteq \mathbb{R}^d$ be an open ball. If $F : B \rightarrow \mathbb{R}^d$ is a C^1 vector field that satisfies the locally conservative condition $\partial_k F_\ell = \partial_\ell F_k$ for all $1 \leq k, \ell \leq d$, then F is conservative in B .

Proof: We will prove this by explicitly constructing a potential function u such that $F = \nabla u$.

Step 1 – The Rectangle Property: First, we consider a rectangle whose sides are parallel to the coordinate axes e_k and e_ℓ . Let the vertices be defined as:



We want to show that integrating (with direction) along the blue path ($A \rightarrow B \rightarrow C$) yields the same result as integrating along the red path ($A \rightarrow D \rightarrow C$).

This is equivalent to showing:

$$\int_{B \rightarrow C} \vec{F} d\vec{t} - \int_{A \rightarrow D} \vec{F} d\vec{t} = \int_{D \rightarrow C} \vec{F} d\vec{t} - \int_{A \rightarrow B} \vec{F} d\vec{t}$$

Let's compute the difference on the left-hand side (the segments parallel to e_ℓ):

$$\int_{B \rightarrow C} - \int_{A \rightarrow D} = \int_0^b (F_\ell(x + ae_k + te_\ell) - F_\ell(x + te_\ell)) dt = \int_0^b \int_0^a (\partial_k F_\ell)(x + se_k + te_\ell) ds dt$$

Similarly, the right-hand side (the segments parallel to e_k) is:

$$\int_{D \rightarrow C} - \int_{A \rightarrow B} = \int_0^a (F_k(x + se_k + be_\ell) - F_k(x + se_k)) ds = \int_0^a \int_0^b (\partial_\ell F_k)(x + se_k + te_\ell) dt ds$$

By our hypothesis, $\partial_k F_\ell = \partial_\ell F_k$.

Applying Fubini's Theorem to swap the order of integration, we see that the two sides are exactly equal.

Hence, the integral over such axis-parallel rectangles is path-independent.

End of Lecture 4 – 16/04

Step 2 – Path Independence for Staircase Paths: Given any two points in the ball, we can connect them using a “staircase” path built by $\gamma = \gamma_1 + \gamma_2 + \dots + \gamma_m$, where each segment γ_i is parallel to one of the coordinate axes.

We claim that the line integral $\int_\gamma \vec{F} d\vec{t}$ is independent of the specific staircase path chosen:

If we have a path with many alternating steps, we can use the rectangle property established in Step 1 to “push” the segments around. For instance, a step in direction e_ℓ followed by a step in e_k can be swapped to a step in e_k followed by e_ℓ without changing the integral's value. By repeating this procedure, we can deform any axis-parallel path into a canonical path (e.g., doing all e_1 movements first, then all e_2 movements, etc.).

Since all such paths can be deformed into the same canonical path without altering the integral, the line integral over axis-parallel paths is well-defined.

Step 3 – Constructing the Potential: Fix a base point $x_0 \in B$. We define our potential function as:

$$u(x) = \int_{x_0 \rightarrow x} \vec{F} d\vec{t}$$

where the integral is taken over any path composed of axis-parallel segments. To verify this is a valid potential, we compute its partial derivative $\partial_k u$:

$$\frac{u(x + he_k) - u(x)}{h} = \frac{1}{h} \left(\int_{x_0 \rightarrow x + he_k} \vec{F} d\vec{t} - \int_{x_0 \rightarrow x} \vec{F} d\vec{t} \right) = \frac{1}{h} \int_{x \rightarrow x + he_k} \vec{F} d\vec{t} = \frac{1}{h} \int_0^h F_k(x + te_k) dt$$

Taking the limit as $h \rightarrow 0$, the Fundamental Theorem of Calculus implies this converges to $F_k(x)$.

Therefore, $\partial_k u = F_k$ for all k , which means $F = \nabla u$. Since F is a gradient field, it is conservative in B . □

End of Lecture 5 – 23/04

Remark

From here until the end of chapter we aim to prove Green's Theorem (Theorem 1.27) and it was spread over lectures 6,7,8. In each part of the proof we used multiple definitions/exercise that each we saw in the matching lecture for the part of the proof we used it so the order is by preference.

Definition 1.21

Locally Conservative

Let $F : \Omega \rightarrow \mathbb{R}^d$ be a vector field. We say that F is **locally conservative** (meshamer mekomit) in Ω if for every $k, \ell = 1, \dots, d$,

$$\partial_k F_\ell = \partial_\ell F_k \iff \frac{\partial F_\ell}{\partial k} = \frac{\partial F_k}{\partial \ell}$$

Definition 1.22

C^k -smooth

Domain $\Omega \subset \mathbb{R}^d$ is **C^k -smooth** if for all $x \in \partial\Omega$ there exists a neighborhood $x \in V$ and a C^k -diffeomorphism $\varphi : B_1(0) \rightarrow V$ such that $\varphi(B_1(0) \cap \{x_d > 0\}) = V \cap \Omega$ and we call φ a rectifying map.

Informal: Ω is smooth if for all $p \in \partial\Omega$ there exists a neighborhood $p \in V$ such that $V \cap \Omega$ looks like a half circle.

Definition 1.23

Simply Connected

X a topological space is **simply connected** if it is a path connected and any loop ($\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = \gamma(1)$) in X can be continuously contracted a point (i.e., is homotopic to a constant loop, with no “holes”).

Theorem 1.24

If Ω is simply connected and F is locally conservative in Ω then F is a conservative in Ω .

Proof: In exercise 3. □

Notation

$$\int_\gamma \vec{F} d\vec{\ell} = \int_\gamma F_1 dx_1 + F_2 dx_2 + \dots + F_\ell dx_\ell$$

Exercise 1.25

Let $p \in \partial\Omega$, V a neighborhood of p and $\varphi : B_1(0) \rightarrow V$ a rectifying map with $\text{supp}(F) \subseteq V$ ($\varphi(t, s) = (x(t, s), y(t, s))$). If $J_p > 0$ then $\varphi(t, 0)$ is a parametrization of $\partial\Omega \cap V$ with positive orientation (since we never use reflection).

Definition 1.26

Partition of Unity

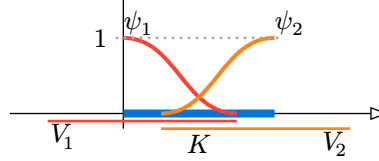
Let $K \subseteq \mathbb{R}^d$ be a compact set and $K \subset V_1 \cup V_2 \cup \dots \cup V_m$ a finite open cover. Then there exists C^k functions $\psi_1, \dots, \psi_m$ defined in a neighborhood of K such that

1. $\forall j, \psi_j \geq 0$
2. $\text{supp}(\psi_j) \subseteq V_k$
3. $\sum_{j=1}^m \psi_j = 1$ on K (meaning for all $x \in K$ we have $\sum_{j=1}^m \psi_j(x) = 1$)

Such collection $\{\psi_j\}_{j=1}^m$ is called **Partition of Unity** on K subordinated to the cover $\{V_j\}_{j=1}^m$.

Example

Let $K = [0, 2]$, $V_1 = (-1, 1)$, $V_2 = (\frac{1}{2}, 3)$ so $K \subset V_1 \cup V_2$ and we define ψ_1, ψ_2



Theorem 1.27

Green's Theorem

Let $\Omega \subset \mathbb{R}^2$ be a bounded smooth domain and Let $F = (P, Q)$ be a smooth vector field in a neighborhood of $\partial\Omega$ and $\partial\Omega$ be positively oriented (left leg rule) then

$$\oint_{\partial\Omega} P dx + Q dy = \oint_{\partial\Omega} \vec{F} d\vec{\ell} = \int_{\Omega} (\partial_x Q - \partial_y P) dx dy$$

Since $\gamma(t) = (x(t), y(t))$, $dx = x'(t) dt$, $dy = y'(t) dt$ we get

$$\int_{\gamma} P dx + Q dy = \int (P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)) dt = \int \left\langle \begin{pmatrix} P \\ Q \end{pmatrix}, \begin{pmatrix} x' \\ y' \end{pmatrix} \right\rangle dt$$

Remark

Green Theorem works for other domains and for example in the exercise we will prove for rectangles.

Example

Suppose that F is locally conservative in $\Omega \subseteq \mathbb{R}^2$ so $F = (P, Q)$ and $\partial_y P = \partial_x Q$ from Green's theorem

$$\int_{\partial\Omega} \vec{F} d\vec{\ell} = 0$$

Proof of Theorem 1.27:

1st case: Suppose $\text{supp}(F) = \overline{\{x \in \Omega \mid F(x) \neq 0\}} \subset \Omega$.

Since $F|_{\partial\Omega} = 0$ we attain that $\int_{\partial\Omega} \vec{F} d\vec{\ell} = 0$ so we need to show that the integral on the right is also zero.

Since Ω is a bounded smooth domain there exists $M \in \mathbb{R}_+$ so $\Omega \subset [-M, M] \times [-M, M]$ hence

$$\begin{aligned} \int_{\Omega} (\partial_x Q - \partial_y P) dx dy &= \int_{[-M, M] \times [-M, M]} (\partial_x Q - \partial_y P) dx dy \\ &= \int_{-M}^M \int_{-M}^M \partial_x Q dx dy - \int_{-M}^M \int_{-M}^M \partial_y P dy dx \\ &= \int_{-M}^M Q(M, y) - Q(-M, y) dy + \int_{-M}^M P(x, M) - P(x, -M) dx = 0 \end{aligned}$$

Where the second equal is Fubini's Theorem and the third one is Fundamental Theorem of Calculus.

2nd case: Assume $\Omega = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1, y > 0\}$ and $\text{supp}(F) \subset B_1(0)$.

First, we need to find a parametrization for our boundary: With $-1 \leq t \leq 1$ we have $\gamma(t) = (t, 0)$ with $\gamma'(t) = (1, 0)$ hence

$$\int_{\partial\Omega} P dx + Q dy = \int_{-1}^1 P(t, 0) \cdot 1 + Q(t, 0) \cdot 0 dt = \int_{-1}^1 P(t, 0) dt$$

We need to compute $\int (\partial_x Q - \partial_y P) dx dy$ over the upper half circle of and show that they are equal.

Over the upper half circle we can replace with rectangle (small enough) and use Fubini's Theorem and Fundamental Theorem of Calculus

$$\int \partial_x Q dx dy = \int_{(-1,1) \times (0,1)} \partial_x Q dx dy = \int_0^1 \int_{-1}^1 \partial_x Q dx dy = \int_0^1 \underbrace{Q(1, y)}_{=0} - \underbrace{Q(-1, y)}_{=0} dy = 0$$

Since we are out of our support. In the same way

$$-\int \partial_y P dx dy = -\int_{(-1,1) \times (0,1)} \partial_y P dx dy = -\int_{-1}^1 \int_0^1 \partial_y P dy dx = -\int_{-1}^1 \underbrace{P(x, 1)}_{=0} - P(x, 0) dx = \int_{-1}^1 P(x, 0) dx$$

Using a change of variable we finish this case.

End of Lecture 6 – 28/04

3rd case: Let $p \in \partial\Omega$, V a neighborhood of p and $\varphi : B_1(0) \rightarrow V$ a rectifying map with $\text{supp}(F) \subseteq V$ ($\varphi(t, s) = (x(t, s), y(t, s))$).

We can assume that $J_\varphi > 0$ (if not, we can replace φ by reflection – simply flipping one coordinate).

Using Exercise 1.25 since $\text{supp}(F) \subseteq V$ we can write with the definition of line integral

$$\int_{\partial\Omega} P dx + Q dy = \int_{\partial\Omega \cap V} P dx + Q dy = \int_{-1}^1 P(\varphi(t, 0)) \frac{\partial x}{\partial t}(t, 0) + Q(\varphi(t, 0)) \frac{\partial y}{\partial t}(t, 0) dt$$

For the other side, we start by computing

$$J_\varphi = \det \begin{pmatrix} \frac{\partial x}{\partial t} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial t} & \frac{\partial y}{\partial s} \end{pmatrix} = \frac{\partial x}{\partial t} \frac{\partial y}{\partial s} - \frac{\partial x}{\partial s} \frac{\partial y}{\partial t}$$

Since F is smooth we use change of variable to get

$$\begin{aligned} \int_{\Omega} \partial_x Q - \partial_y P dx dy &= \int_{\Omega \cap V} \partial_x Q - \partial_y P dx dy \\ &= \int_{\mathbf{A}} (\partial_x Q \circ \varphi - \partial_y P \circ \varphi) J_\varphi dt ds \\ &= \int_{\mathbf{A}} (\partial_x Q \circ \varphi - \partial_y P \circ \varphi) \left(\frac{\partial x}{\partial t} \frac{\partial y}{\partial s} - \frac{\partial x}{\partial s} \frac{\partial y}{\partial t} \right) dt ds \\ &= \int_{\mathbf{A}} \left((\partial_x Q \circ \varphi) \frac{\partial x}{\partial t} + (\partial_y Q \circ \varphi) \frac{\partial y}{\partial t} \right) \frac{\partial y}{\partial s} - \left((\partial_x Q \circ \varphi) \frac{\partial x}{\partial s} + (\partial_y Q \circ \varphi) \frac{\partial y}{\partial s} \right) \frac{\partial y}{\partial t} \\ &\quad + \left((\partial_x P \circ \varphi) \frac{\partial x}{\partial t} + (\partial_y P \circ \varphi) \frac{\partial y}{\partial t} \right) \frac{\partial x}{\partial s} - \left((\partial_x P \circ \varphi) \frac{\partial x}{\partial s} + (\partial_y P \circ \varphi) \frac{\partial y}{\partial s} \right) \frac{\partial x}{\partial t} \\ &\stackrel{(1)}{=} \int_{\mathbf{A}} \left[\frac{\partial(Q \circ \varphi)}{\partial t} \frac{\partial y}{\partial s} - \frac{\partial(Q \circ \varphi)}{\partial s} \frac{\partial y}{\partial t} + \frac{\partial(P \circ \varphi)}{\partial t} \frac{\partial x}{\partial s} - \frac{\partial(P \circ \varphi)}{\partial s} \frac{\partial x}{\partial t} \right] dt ds \\ &\stackrel{(2)}{=} \int_{\mathbf{A}} \left[\frac{\partial}{\partial t} \left((Q \circ \varphi) \frac{\partial y}{\partial s} \right) - \frac{\partial}{\partial s} \left((Q \circ \varphi) \frac{\partial y}{\partial t} \right) + \frac{\partial}{\partial t} \left((P \circ \varphi) \frac{\partial x}{\partial s} \right) - \frac{\partial}{\partial s} \left((P \circ \varphi) \frac{\partial x}{\partial t} \right) \right] dt ds \\ &= \int_{\mathbf{A}} \left[\frac{\partial}{\partial t} \left(\underbrace{(P \circ \varphi) \frac{\partial x}{\partial s} + (Q \circ \varphi) \frac{\partial y}{\partial s}}_{\tilde{Q}} \right) - \frac{\partial}{\partial s} \left(\underbrace{(P \circ \varphi) \frac{\partial x}{\partial t} + (Q \circ \varphi) \frac{\partial y}{\partial t}}_{\tilde{P}} \right) \right] dt ds \end{aligned}$$

Where (1) is Chain Rule and (2) Leibniz Integral Rule (Notice the line break with the +).

If we take $\vec{G}(t, s) = (\tilde{P}, \tilde{Q})$ and since $\gamma(t) = (t, 0)$, $\dot{\gamma}(t) = (1, 0)$, the 2nd case gives us

$$\int_{\partial\Omega} \tilde{P} dt + \tilde{Q} ds = \int_{\mathbf{a}} \frac{\partial \tilde{Q}}{\partial t} - \frac{\partial \tilde{P}}{\partial s} dt ds \implies \int \tilde{P} dt + \tilde{Q} ds = \int_{-1}^1 \tilde{P} dt = \int_{-1}^1 \left[(P \circ \varphi) \frac{\partial y}{\partial t}(t, 0) + (Q \circ \varphi) \frac{\partial y}{\partial t}(t, 0) \right] dt ds$$

Which is indeed the left side.

In other words for the 3rd case: we pull our vector field back to the upper half circle and go back to the 2nd case.

End of Lecture 7 – 30/04

4rd case: For all $p \in \partial\Omega$ there exists a neighborhood V_p and a rectifying map $\varphi_p : B_1(0) \rightarrow V_p$.

Since $\partial\Omega \subset \bigcup_{p \in \partial\Omega} V_p$ and since $\partial\Omega$ is compact then there exists a finite open subcover $\partial\Omega \subset V_{p_1} \cup \dots \cup V_{p_m}$. Let $V_0 = \Omega$ and $\bar{\Omega} \subset V_0 \cup V_{p_1} \cup \dots \cup V_{p_m}$ and for simplicity of notations let us write $v_j = v_{\gamma_j}$.

There exists a partition of unity on $\bar{\Omega}$ subordinate to $\{V_j\}_{j=0}^m$ and we define $\vec{F}_j := \psi_j \cdot \vec{F}$ and $\text{supp}(\vec{F}_j) \subset V_j$ so we have

$$\sum_{j=1}^m \vec{F}_j = \sum_{j=1}^m (\psi_j \cdot \vec{F}) = \left(\sum_{j=1}^m \psi_j \right) \cdot \vec{F} \underset{\text{on } \bar{\Omega}}{=} 1 \cdot \vec{F} = \vec{F}$$

From the 1st case if we set $\vec{F}_j = (P_j, Q_j)$ since $\text{supp}(\vec{F}_0) \subseteq \Omega$ we have that

$$\int_{\partial\Omega} \vec{F}_0 d\vec{\ell} = \int_{\gamma} \partial_X Q_0 - \partial_y P_0 dx dy$$

By the 3rd case we know that for each $1 \leq j \leq m$ we have

$$\int_{\partial\Omega} \vec{F}_j d\vec{\ell} = \int_{\gamma} \partial_x Q_j - \partial_y P_j dx dy$$

Summation gives us

$$\sum_{j=0}^m \int_{\partial\Omega} \vec{F}_j d\vec{\ell} = \sum_{j=0}^m \int_{\gamma} \partial_x Q_j - \partial_y P_j dx dy \implies \int_{\partial\Omega} \vec{F} d\vec{\ell} = \int_{\gamma} \partial_x Q - \partial_y P dx dy$$

□

Proof of the existence of partition of unity from the notes by Or HersHKovits, Theorem 10.3:

It will be easier to work with balls rather than a general covering.

Since each U_i is open in K , there exists an open set $\tilde{U}_i \subseteq \mathbb{R}^n$ such that $U_i = \tilde{U}_i \cap K$ and for every $p \in K$, there exists a ball $B(p, r_p)$ such that $B(p, 4r_p) \subseteq \tilde{U}_i$ for some i .

The collection $\{B(p, r_p)\}$ is an open covering of K , and since K is compact, there exist finitely many points $\{p_j\}_{j=1}^T$ such that $\{B(p_j, r_j)\}_{j=1}^T$ is a finite subcover of K so for each j , consider the function $\psi_j : \mathbb{R}^n \rightarrow \mathbb{R}$ given by:

$$\psi_j(x) = \mu \left(\frac{x - p_j}{r_j} \right)$$

Note that $\psi_j > 0$ on $B(p_j, r_j)$ and the support of ψ_j is a compact subset of $\tilde{U}_i$ (since $B(p_j, 4r_j) \subseteq \tilde{U}_i$). For every $i = 1, \dots, m$, set:

$$\tilde{\phi}_i = \sum_{\{j \mid B(p_j, 4r_j) \subseteq \tilde{U}_i\}} \psi_j$$

So the support of $\tilde{\phi}_i$ is compact in $\tilde{U}_i$. Define $V = \bigcup_{j=1}^T B(p_j, r_j)$ and we have that V is an open set satisfying $K \subseteq V \subseteq \bigcup \tilde{U}_i$ and the balls cover V so we know that

$$\sum_{i=1}^m \tilde{\phi}_i(x) > 0 \quad \text{for all } x \in V$$

Finally, we normalize these functions to construct the partition of unity. For $x \in V$, we define:

$$\phi_i(x) = \frac{\tilde{\phi}_i(x)}{\sum_{k=1}^m \tilde{\phi}_k(x)}$$

Restricting each ϕ_i to K yields the desired partition of unity subordinate to $\{U_i\}$.

□

2 Smooth Manifolds in $\mathbb{R}^d$

Definition 2.1

Smooth manifolds in $\mathbb{R}^d$

Let $k \leq d$, a set $M \subseteq \mathbb{R}^d$ is called a **manifold** of dimension k if for all $p \in M$ there exists $W \subseteq \mathbb{R}^d$ a neighborhood of p ($p \in W$), an open set $U \subseteq \mathbb{R}^k$ and a smooth map $\varphi : U \rightarrow W$ so that

1. φ is a homeomorphism on its image
2. $\varphi(U) = M \cap W$
3. $\text{rank}(D\varphi_u) = k$ for all $u \in U$

Informly, a manifold of dimension k is any topological space U so that any point $p \in U$ has a neighborhood W which is a homeomorphism to a ball in $\mathbb{R}^k$ (A k -dimensional manifold is a space that locally looks like $\mathbb{R}^k$).

End of Lecture 8 – 05/05

TBD – Lecture 9

End of Lecture 9 – 07/05

Definition 2.2

Graph of dim k

A set $\Gamma \subseteq \mathbb{R}^d$ is a **graph of dim k** if there exists a partition $\{1, 2, \dots, d\} = I \cup J$ with $|I| = k$, an open set $U \subseteq \mathbb{R}^k$ and a function $f : U \rightarrow \mathbb{R}^{d-k}$ such that

$$\Gamma = \{x \in \mathbb{R}^d \mid x_I \in U, x_J = f(x_I)\}$$

Example

Let $U \subseteq \mathbb{R}^d$ be an open set with $k = d$. We show that U is a graph.

We take $I = \{1, 2, \dots, d\}$ and $J = \emptyset$ with $f : U \rightarrow \mathbb{R}^0 = \{0\}$ defined by $f(u) = 0$.

Then

$$U = \{x \in \mathbb{R}^d \mid x \in U\} = \{x \in \mathbb{R}^d \mid x_I \in U, x_J = f(x_I)\}$$

Since $J = \emptyset$, the condition $x_J = f(x_I)$ is vacuous.

Definition 2.3

Smooth manifolds of dim k by graph

A set $M \subseteq \mathbb{R}^d$ is a **smooth manifold of dim k** if for every $p \in M$ there exists an open neighborhood $W \subseteq \mathbb{R}^d$ of p such that $W \cap M$ is a graph of dim k of a smooth function. We say that M is locally a smooth graph of dim k for a $p \in M$.

Definition 2.4

Manifold as the null set of a system of equations

Let $M \subseteq \mathbb{R}^n$. We say that M is a k dimensional manifold if for all $p \in M$ there exists a neighborhood $W \subseteq \mathbb{R}^n$ and a smooth function $F : W \rightarrow \mathbb{R}^{n-k}$ such that

1. $M \cap W = \{x \in W \mid F(x) = 0\}$
2. $DF_p : \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$ is of full rank

Definition 2.5

Smooth manifolds of dim k

Let $k \leq d$. A set $M \subseteq \mathbb{R}^d$ is called a **smooth manifold of dim k** if for every $p \in M$ there exists an open neighborhood $W \subseteq \mathbb{R}^d$ of p , an open set $U \subseteq \mathbb{R}^k$ and a smooth map $\varphi : U \rightarrow W$ such that

1. $\varphi : U \rightarrow \varphi(U)$ is a homeomorphism
2. $\varphi(U) = M \cap W$
3. $\text{rank}(D\varphi_u) = k$ for all $u \in U$

Example

We look at the curve

$$\gamma(t) = (\sin(t), \sin(2t))$$

We define $S = \{\gamma(t) \mid 0 < t < \frac{5\pi}{4}\}$ and ask is S a smooth manifold according to the definition with graphs? Observe that

$$\gamma(0) = \gamma(\pi) = (0, 0)$$

hence the parametrization is not injective.

On the other hand, define $U = (-\varepsilon, \frac{5\pi}{4})$ then

$$(D\varphi)|_{t=0} = (\cos(t), 2\cos(2t))|_{t=0} = (1, 2)$$

has full rank.

However, since $\varphi(0) = \varphi(\pi)$, the map φ is not injective, hence it is not a homeomorphism onto its image.

Therefore this parametrization does not define a smooth manifold chart.

In fact, near $(0, 0)$ the set looks like two curves crossing, so S is not a smooth manifold.

Proposition 2.6

Every smooth manifold by the graph definition is a smooth manifold.

Proof: Let

$$\Gamma = \{x \in \mathbb{R}^d \mid x_I \in U, x_J = f(x_I)\}$$

where $f : U \rightarrow \mathbb{R}^{d-k}$, $\{1, 2, \dots, d\} = I \cup J$ and $|I| = k$.

Define $\varphi : U \rightarrow \Gamma$ by

$$(\pi_I \circ \varphi)(u) = u, \quad (\pi_J \circ \varphi)(u) = f(u)$$

The map φ is smooth since each coordinate is a smooth function.

Moreover, the Jacobian matrix of φ contains the identity matrix I_k as a block, hence $\text{rank}(D\varphi_u) = k$ for all $u \in U$.

Hence

$$\varphi(U) = \{\varphi(u) \mid u \in U\} = \{x \in \mathbb{R}^d \mid (\pi_I \circ \varphi)(x) = u, (\pi_J \circ \varphi)(x) = f(u)\}$$

We have $\varphi : U \rightarrow \Gamma$ invertible since

$$(\pi_I \circ \varphi)(u) = \pi_I(\varphi(u)) = u$$

Moreover,

$$\pi_I(\varphi((\pi_I \circ \varphi)^{-1}(\pi_I(x)))) = \pi_I(x), \quad \pi_J(\varphi((\pi_I \circ \varphi)^{-1}(\pi_I(x)))) = f(\pi_I(x)) = \pi_J(x)$$

Therefore

$$\varphi((\pi_I \circ \varphi)^{-1}(\pi_I(x))) = x$$

So $\pi_I : \Gamma \rightarrow U$ is continuous and is the inverse of $\varphi : U \rightarrow \Gamma$. □

Proposition 2.7

Every smooth manifold is a graph manifold.

Proof: Let $p \in M$, $p \in W \subseteq \mathbb{R}^d$ open, $U \subseteq \mathbb{R}^k$ and $\varphi : U \rightarrow W$ with $\varphi(U) = W \cap M$, such that $(D\varphi)|_{\varphi^{-1}(p)}$ is of full rank.

There exists $I \subseteq \{1, 2, \dots, d\}$ with $|I| = k$ so that $(D\varphi)|_{I \times \{1, 2, \dots, k\}}$ is invertible.

We look at the map $\pi_I \circ \varphi : U \rightarrow \mathbb{R}^k$ which is a smooth map from U to $\mathbb{R}^k$ and

$$D(\pi_I \circ \varphi)|_{\varphi^{-1}(p)} = (D\varphi)|_{I \times \{1, 2, \dots, k\}}$$

Is invertible.

By the inverse function theorem, $\pi_I \circ \varphi$ is a local diffeomorphism at $\varphi^{-1}(p)$, so there exists $\varphi^{-1}(p) \in U_0 \subset U$ open and there exists $V_0 \subseteq \mathbb{R}^k$ open so that $(\pi_I \circ \varphi) : U_0 \rightarrow V_0$ is a diffeomorphism.

We define $f : V_0 \rightarrow \mathbb{R}^{d-k}$ by

$$f(v) = \pi_J(\varphi((\pi_I \circ \varphi)^{-1}(v)))$$

Where $J = \{1, 2, \dots, d\} \setminus I$. Let W_0 be such that $\varphi(U_0) = W_0 \cap M$.

We need to verify that

$$\underbrace{W_0 \cap M}_{=A} = \underbrace{\{x \in \mathbb{R}^d \mid \pi_I(x) \in V_0, \pi_J(x) = f(\pi_I(x))\}}_{=B}$$

Since $W_0 \cap M = \varphi(U_0)$, let $u \in U_0$.

Then there exists $v \in V_0$ such that $(\pi_I \circ \varphi)^{-1}(v) = u$, we aim to show that $\varphi(u) \in B$.

From our definition, $\pi_I(\varphi(u)) = v \in V_0$.

Also,

$$\pi_J(\varphi(u)) = \pi_J(\varphi((\pi_I \circ \varphi)^{-1}(v))) = f(v)$$

Hence $\varphi(u) \in B$.

Why is $B \subseteq A$? Let $x \in B$ and define

$$u = (\pi_I \circ \varphi)^{-1}(\pi_I(x)) \in U_0$$

Then

$$\pi_I(\varphi(u)) = (\pi_I \circ \varphi)(u) = \pi_I(x)$$

Also,

$$\pi_J(\varphi(u)) = f(\pi_I(\varphi(u))) = f(\pi_I(x)) = \pi_J(x)$$

Because the coordinates agree on both I and J , we conclude that $\varphi(u) = x$, hence $x \in A$. □

Proposition 2.8

Every smooth manifold by the graph definition is manifold by the null set of a system of equations definition (and by propositions above, also by the smooth manifold definition)

Proof: In Recitation 5 + Exercise 5. □

2.1 Transition Maps

Let M be a smooth manifold of dim k and suppose we have

$$\varphi_1 : U_1 \rightarrow W_1, \quad \varphi_2 : U_2 \rightarrow W_2$$

where W_1, W_2 are neighborhoods of p .

Denote

$$\tilde{U}_1 = \varphi_1^{-1}(W_1 \cap W_2 \cap M), \quad \tilde{U}_2 = \varphi_2^{-1}(W_1 \cap W_2 \cap M)$$

We define $\tau_{12} : \tilde{U}_1 \rightarrow \tilde{U}_2$ by

$$\tau_{12}(u) = \varphi_2^{-1}(\varphi_1(u))$$

where $\varphi_2 : U_2 \rightarrow W_2 \cap M$.

We ask ourselves: is τ_{12} a smooth map?

After possibly shrinking neighborhoods, we may assume both charts are represented as graphs over the same coordinate subset I .

Then

$$\tau_{12} = \varphi_2^{-1} \circ \varphi_1 = (\pi_I \circ \varphi_2)^{-1} \circ (\pi_I \circ \varphi_1)$$

Since compositions and inverses of smooth maps are smooth, the transition map is smooth.

End of Lecture 10 – 12/05

2.2 Integrals On Parametrized Manifolds

Our goal is to define $\int_M f \, d \text{Vol}_k$ where $M \subseteq \mathbb{R}^d$ a k -dim manifold. **TBD.**